

DSBD 2

Code:

```
CREATE DATABASE IF NOT EXISTS week2;
USE week2;

-- BOOKS TABLE
CREATE TABLE Books (
    book_id INT PRIMARY KEY,
    title VARCHAR(100) NOT NULL,
    isbn VARCHAR(20) UNIQUE,
    published_year INT CHECK (published_year < 2027)
);

INSERT INTO Books (book_id, title, isbn, published_year) VALUES
    (1, 'The Great Gatsby', '9780743273565', 1925),
    (2, 'To Kill a Mockingbird', '9780061120084', 1960),
    (3, '1984', '9780451524935', 1949);

SELECT * FROM Books;

-- MEMBERS TABLE
CREATE TABLE Members (
    member_id INT PRIMARY KEY,
    full_name VARCHAR(100),
    email VARCHAR(100) UNIQUE
);

INSERT INTO Members (member_id, full_name, email) VALUES
    (101, 'Kankshitha', 'kankshitha@email.com'),
    (102, 'Chaitanya', 'chaitanya@email.com'),
    (103, 'Karthik', 'karthik@email.com');

SELECT * FROM Members;

-- LOANS TABLE
CREATE TABLE Loans (
    loan_id INT PRIMARY KEY,
    member_id INT,
    book_id INT,
    loan_date DATE,
    FOREIGN KEY (member_id) REFERENCES Members(member_id),
    FOREIGN KEY (book_id) REFERENCES Books(book_id)
);

INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
    (1, 101, 1, '2025-01-05'),
```

```
(2, 102, 2, '2025-01-08'),
(3, 103, 3, '2025-01-10'),
(4, 101, 2, '2025-02-01'),
(5, 102, 1, '2025-02-05'),
(6, 103, 2, '2025-02-12'),
(7, 101, 3, '2025-03-01'),
(8, 102, 3, '2025-03-07'),
(9, 103, 1, '2025-03-15'),
(10, 101, 1, '2025-04-01');
```

```
SELECT * FROM Loans;
```

```
-- JOIN QUERY
```

```
SELECT
```

```
    m.full_name AS Member_Name,
    b.title AS Book_Title
```

```
FROM Loans l
```

```
    INNER JOIN Members m ON l.member_id = m.member_id
    INNER JOIN Books b ON l.book_id = b.book_id;
```

```
-- GROUP BY QUERY
```

```
SELECT
```

```
    published_year,
    COUNT(book_id) AS Total_Books
```

```
FROM Books
```

```
GROUP BY published_year
```

```
ORDER BY published_year;
```

```
-- DONATION HISTORY TABLE
```

```
CREATE TABLE Donation_History (
```

```
    donation_id INT PRIMARY KEY,
    book_id INT,
    donor_name VARCHAR(100),
    donation_date DATE,
    FOREIGN KEY (book_id) REFERENCES Books(book_id)
```

```
);
```

```
-- TRANSACTION
```

```
START TRANSACTION;
```

```
INSERT INTO Books (book_id, title, isbn, published_year)
VALUES (4, 'Animal Farm', '9780451526342', 1945);
```

```
INSERT INTO Donation_History
(donation_id, book_id, donor_name, donation_date)
VALUES (1, 4, 'Raj Kumar', CURDATE());
```

```
COMMIT;
```

```
-- INDEX
CREATE INDEX idx_books_isbn ON Books(isbn);

-- TEST QUERY
SELECT * FROM Books
WHERE isbn = '9780451524935';
```

Output:

Output week2.Loans				
	loan_id	member_id	book_id	loan_date
1		1	101	2025-01-05
2		2	102	2025-01-08
3		3	103	2025-01-10
4		4	101	2025-02-01
5		5	102	2025-02-05
6		6	103	2025-02-12

Output week2.Members			
	member_id	full_name	email
1	101	Kankshitha	kankshitha@email.com
2	102	Chaitanya	chaitanya@email.com
3	103	Karthik	karthik@email.com

Output TEST QUERY				
	book_id	title	isbn	published_year
1	3	1984	9780451524935	1946